

Reproducing Paper 1: Attention at Theoretical Minimum — MoA DNF with Portable Code

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Abstract

Reproduction of Paper 1 arXiv:2606.07713 from dissertation 1988 full psi and Omega through to verified portable code only OpenMP, OpenACC, OpenMPI and deterministic flags fitting T approx alpha F + beta M. Exact inputs Sec 5.2 verified Tables 1,2 to 1e-4. Appendix contains full executable Python and C/OpenMP, Fortran/OpenACC, MPI.

1 Dissertation Foundation

Full psi partial indexing rho(i psi xi)=tau i drop rho xi, take v uparrow xi Def 1.0, drop Def 1.1, reverse phi, rotate theta, transpose oslash, Lemmas L1-L5 distributing psi over catenation and unary ops. Gamma Eq3: gamma(v,s)=sum v_k prods_j row - major.Omegasubset : +_Omega < 0, 1 >, x_Omega < 0, 1 >, +_Omega < 1, 1 >, max_Omega < 1 > .

2 Paper 1 DNF

Eq8 Attention=softmax(QK^T/sqrt(dk))V, rho =< 3, 4 > Eq9.StepIEq13z = x(-Omega < 1, 0 >)((ceilred_Omega < 1 >)x), psi - reductionEq14 < i'' > psiz =< i'' > psix - ceilred(< i'' > psix).StepIIEq15 - 17num = e^z, < i'', k > psi(Q + .xK^T) = red_+(< i'', j > psiQx < k, j > psiK)eliminatesK^Tbuffer.StepIIIEq18 - 19den = (red_+Omega < 1 >)e^z, < i'' > psiden = red_+(< i'' > psie^z).StepIVY = num/den.CompleteDNFEq20, OutputEq21 < i'' > psiOutput = red_+(< i'', p > psiYx < p > psiV).Theorem3.1StorageTheoremminimal.Prop4.1M_classical = O(n^2 + nd_k + nd_v)Eq23, M_MoA = O(nd_k + nd_v)Eq24, kapproxn/dkEq25512xatn = 32768.

3 Exact Verification

Inputs Q,K,V from Sec 5.2 reproduced Tables 1,2 to 1e-4 checkmark. Trace j0_i psi Output via Eq17-21. ONF gamma Eq7 gamma((0,0),j3,4_i)=0, gamma((0,3))=3, gamma((1,2))=6, gamma((2,1))=9.

4 Portable Deterministic Emission

From hetero_m ainandclosing_p rediction_g ap : 5-deviceclosurereached(d)M1exp, Pd, Xd = 0percent, Cd, Rd0.00028measuredviabracet-and-locatesweepLevelZeroSysman99.88percentbusy.OnlyOpenMP, OpenACC, O2 - fno - fast - math - fopenmp - fopenacc - acc - Minfo = accel - ta = tesla : cc80 - DOMP_DETERMINISTIC-DOMPI_DETERMINISTICforbidden-O3-Ofast.Eachpragmamaps1-1toONFstart/stop/stridesoTapproxalphaF+betaMbeta >> alpha100-1000xDRAMvsFLOPpredictionmat 2609.33916, 535xNUMAvs < 3x, CvsFortran3.17x timematches3.35xstallsgammarowvscol.

A Full Executable Code Paper 1 Exact

```
import numpy as np

print("====Paper 1 Reproduction: Attention at Theoretical Minimum====")
print("From arXiv:2606.07713, Sec 4.3-5.3, Listing 1")
print()

Q = np.array([
    [-0.1984, 0.2698, 0.3414, -0.0372],
    [ 0.2547, -1.0674, 0.3460, -2.5242],
    [ 0.6822, -0.6265, 0.0252, 0.3978]
], dtype=np.float64)

K = np.array([
    [-1.1567, 0.6885, -0.1884, 0.4743],
    [ 0.2246, 1.7564, 0.5235, -2.3014],
    [-1.5899, 0.3730, -0.8257, -1.2069]
], dtype=np.float64)

V = np.array([
    [ 1.0739, 0.4006, -0.9671, 0.4870],
    [ 0.5589, -0.7209, -0.7650, 0.2689],
    [ 0.8237, 0.3763, 0.8320, 0.0014]
], dtype=np.float64)

n, dk = Q.shape
_, dv = V.shape
print(f"rho(Q)=rho(K)=rho(V)=<{n},{dk}> as in Eq (9)")

def gamma(v, s):
    off=0
    for k in range(len(v)):
        prod=1
        for j in range(k+1, len(s)):
            prod*=s[j]
        off+=v[k]*prod
    return off

def stable_softmax(x, dim=-1):
    z = x - np.max(x, axis=dim, keepdims=True)
    num = np.exp(z)
    den = np.sum(num, axis=dim, keepdims=True)
    return num / den

scores = (Q @ K.T) / np.sqrt(dk)
print("scores = Q K^T / sqrt(dk) =")
print(scores)
print()
```

```

attn_weights = stable_softmax(scores, dim=-1)
print(f"attn_weights_{Y_rho}=<{n},{n}>_{DNF_Eq}(20)")
print(attn_weights)
print()

output = attn_weights @ V
print(f"Output_rho=<{n},{dv}>_{Eq}(21)")
print(output)
print()

expected_weights = np.array([
    [0.3202, 0.3834, 0.2964],
    [0.0288, 0.7300, 0.2412],
    [0.4270, 0.2844, 0.2887]
])
expected_output = np.array([
    [0.8023, -0.0366, -0.3563, 0.2595],
    [0.6376, -0.4240, -0.3857, 0.2107],
    [0.8552, 0.0747, -0.3903, 0.2848]
])

print("==_Verification_vs_Table_1_(weights)_==")
for i in range(n):
    match = np.allclose(attn_weights[i], expected_weights[i], atol=1e-4)
    print(f"Row_{i}:_{attn_weights[i]}_vs_{expected_weights[i]}_Match={match}_{' ' if match else 'not match'}")
print()

print("==_Verification_vs_Table_2_(output)_==")
for i in range(n):
    match = np.allclose(output[i], expected_output[i], atol=1e-4)
    print(f"Row_{i}:_{output[i]}_vs_{expected_output[i]}_Match={match}_{' ' if match else 'not match'}")
print()

print("==_MoA_DNF_psi-reduction_trace_for_{<0>_psi_Output}_==")
i_pp = 0
row0_scores = []
for k in range(n):
    s = 0.0
    for j in range(dk):
        s += Q[i_pp, j] * K[k, j]
    s /= np.sqrt(dk)
    row0_scores.append(s)
row0_scores = np.array(row0_scores)
print(f"<{i_pp},k>_psi_{(Q+.x.K^T)/sqrt}_{row0_scores}")
m = np.max(row0_scores)
z = row0_scores - m
print(f"<{i_pp}>_psi_z_{x}_{max}_{z}_{Eq}(14)")
num = np.exp(z)
print(f"<{i_pp}>_psi_num_{exp(z)}_{num}_{Eq}(16)")
den = np.sum(num)
print(f"<{i_pp}>_psi_den_{red_+(num)}_{den}_{Eq}(19)")

```

```

Y_row = num / den
print (f"<{i_pp}>_psi_Y={num/den}={Y_row}_Eq_(20)")
out_row = np.zeros(dv)
for p in range(n):
    out_row += Y_row[p] * V[p]
print (f"<{i_pp}>_psi_Output={red_+(Y_x_V)}={out_row}_Eq_(21)")
print (f"Matches_output_row_0: {output[0]}")
print ()

print ("==_Prop_4.1_Memory_Minimality_==")
M_classical = n*n + n*dk + n*dk + n*n + n + n*n + n*dv
M_moa = n*dk + n*dk + n*dv
print (f"M_classical_O(n^2+nd_k+nd_v)={M_classical}_Eq_(23)")
print (f"M_moa_O(nd_k+nd_v)={M_moa}_Eq_(24)_reduction_{M_classical/M_moa:.1f}x")
print ()

print ("==_ONF_gamma_Eq_(7)_==")
for idx in [(0,0),(0,3),(1,2),(2,1)]:
    off = gamma(idx, (n,dv))
    print (f"gamma({idx},{n},{dv})={off}")
print ()
print ("==_Paper_1_reproduction_SUCCESS_==")

```

B Compiler Core

”

MoA_Verified_Portable_Compiler_ from_Dissertation_1988_to_Papers_I-V+_Predictions_Only_OpenMP,_OpenACC,_OpenMPI_ deterministic_flags_fitting_ T F + M

```

import numpy as np
from pathlib import Path

```

```

class MoACompiler:
    def __init__(self):
        self.dnf_cache = {}

    def psi_reduce(self, Q, K, V):
        ”Full_psi-reduction_Eq_(17)_eliminates_K^T_buffer,_no_intermediate_DRAM”
        n, dk = Q.shape
        _, dv = V.shape
        # DNF: <i'',k> psi (Q +.x K^T) = red_+(<i'',j> psi Q x <k,j> psi K)
        # Deterministic, no fast-math
        scores = np.zeros((n,n), dtype=np.float64)
        for i in range(n):
            for k in range(n):
                s = 0.0
                for j in range(dk): # gamma: offset = i*dk+j and k*dk+j
                    s += Q[i,j] * K[k,j]
                scores[i,k] = s / np.sqrt(dk)

```

```

# Eq (14)–(20) stable softmax via  $\Omega_{<1,0>}$  and  $\Omega_{<1>}$ 
max_rows = np.max(scores, axis=1, keepdims=True) # red ceil  $\Omega_{<1>}$ 
z = scores - max_rows #  $-\Omega_{<1,0>}$ 
num = np.exp(z) # Eq 16
den = np.sum(num, axis=1, keepdims=True) # red+  $\Omega_{<1>}$  Eq 19
Y = num / den # Eq 20
# Eq 21:  $\langle i'' \rangle$  psi Output = red+ ( $\langle i'' \rangle, p$ ) psi  $Y x \langle p \rangle$  psi  $V$ 
O = Y @ V
return O, scores, Y, {"M1exp": n*dk+n*dk+n*dv, "Pd": n*n, "Xd": 0, "Cd":

def emit_c_openmp(self, n=3, dk=4, dv=4):
    "Emit_C_OpenMP_deterministic _schedule(static)_proc_bind(close)"
    return "
// MoA_DNF->ONF via gamma Eq (7) _C_OpenMP_portable, _deterministic_flags:_
// rho(Q)=rho(K)=rho(V)=<3,4> Eq (9), O(nd_k+nd_v) Prop 4.1
#include <math.h>
#pragma omp parallel_for_schedule(static)_proc_bind(close) //_deterministic, _fit
for_(int i_pp=0; i_pp<3; i_pp++){ //_<i''> psi Output
    double q_i[4];
    for_(int j=0; j<4; j++){ q_i[j] = Q[i_pp*4+j]; //_<i'',j> psi Q, gamma=i_pp*4
    double scores[3];
    for_(int k=0; k<3; k++){ //_<k,j> psi K_direct, no_K^T buffer Eq (17)
        double s=0.0;
        for_(int j=0; j<4; j++){ //_red+_deterministic, no_ffast-math
            s += q_i[j] * K[k*4+j]; //_gamma=k*4+j
        }
        scores[k] = s / sqrt(4.0);
    }
    double max_row = scores[0];
    for_(int k=1; k<3; k++){ if (scores[k]>max_row) max_row=scores[k]; //_ceil_red
    double num[3], den=0.0;
    for_(int k=0; k<3; k++){ _num[k]=exp(scores[k]-max_row); _den+=num[k]; } //_Eq
    double Y[3];
    for_(int k=0; k<3; k++){ Y[k]=num[k]/den; //_Eq 20
    for_(int jv=0; jv<4; jv++){ //_Eq 21
        double acc=0.0;
        for_(int p=0; p<3; p++){ acc += Y[p]*V[p*4+jv];
        Output[i_pp*4+jv]=acc; //_gamma Eq (7)
    }
}
"

def emit_fortran_openacc(self):
    "Emit_Fortran_OpenACC _vector_length(64)_deterministic_as_in_closing_
    return "
! MoA_DNF->ONF Fortran_OpenACC _deterministic_flags:_acc_Minfo=accel-ta=
! _vector_length(64)_avoided_heuristic, _128_on_A100_as_per_closing_paper_Table_X
! $acc_parallel_loop_gang_worker_vector_length(64)_independent
do i_pp=1,3
    !<i'',j> psi Q
    q_i = Q(i_pp,:)

```

```

do k=1,3
  s=0.0d0
  do j=1,4
    s=s+q_i(j)*K(k,j)!_No_K^T_buffer_Eq_(17)
  end do
  scores(k)=s/sqrt(4.0d0)
end do
max_row=maxval(scores)!_ceil_Omega<1>
num=exp(scores-max_row)!_Eq_16
den=sum(num)!_red+_Eq_19
Y=num/den!_Eq_20
do jv=1,4
  Output(i_pp,jv)=sum(Y(:)*V(:,jv))!_Eq_21_red+_
end do
end do
!$acc_end_parallel

```

```

def emit_mpi(self):
    "Emit_OpenMPI_weighted_comm-free_partition_from_hetero_main"
    return "
//_MoA_MPI_weighted_comm-free_partition_hetero_main:_rho_machine(d)=<n0_#_
//_K_new=_K_old_#_k_t_via_MPI_Scatterv/Gatherv,_X_d=0%_construction
MPI_Scatterv(Q_global,_sendcounts,_displs,_MPLDOUBLE,_Q_local,_n_local*dk,_MPI
#pragma omp parallel for schedule(static)
for_(int_i=0;i<n_local;i++)_{
  _//_same_DNF_as_above,_local_O(nd_k+nd_v)
}
MPI_Gatherv(Output_local,_n_local*dv,_MPLDOUBLE,_Output_global,_recvcounts,_dis
//_reached(d)_measured_via_Level_Zero_Sysman_99.88%_busy_Sec_XII_closing_paper

```

```

compiler = MoACompiler()
Q = np.array([[ -0.1984,0.2698,0.3414, -0.0372],[0.2547, -1.0674,0.3460, -2.5242],[0
K = np.array([[ -1.1567,0.6885, -0.1884,0.4743],[0.2246,1.7564,0.5235, -2.3014],[ -1
V = np.array([[1.0739,0.4006, -0.9671,0.4870],[0.5589, -0.7209, -0.7650,0.2689],[0.

```

```

O, scores, Y, metrics = compiler.psi_reduce(Q,K,V)
print("MoA_Compiler_DNF_Output:")
print(O)
print(" Metrics_Mlexp,Pd,Xd,Cd,Rd:", metrics)
print()
print("——_C_OpenMP_Emission——")
print(compiler.emit_c_openmp())
print("——_Fortran_OpenACC_Emission——")
print(compiler.emit_fortran_openacc())
print("——_MPI_Emission——")
print(compiler.emit_mpi())

```